# ASSIGNMENT – 1

Foundations of Cryptography | BCA | Course: BCA309T

### Q1. Define cryptography and its three main goals.

Cryptography protects information by converting it into a secure form. Its main goals are confidentiality, integrity, and authentication; non-repudiation is also an important goal.

### Q2. Differentiate between plaintext and ciphertext.

Plaintext is the original readable message, while ciphertext is the encrypted, unreadable form. Example: “HELLO” may become “KHOOR” using a Caesar cipher.

### Q3. Explain cryptography and why it is essential today. Give three applications.

Cryptography is the method of protecting information using mathematical techniques and keys. It helps prevent unauthorized people from reading or changing data. It is essential for online banking, secure messaging apps, and e-commerce payments. It is also used to protect passwords, files, and online communications.

### Q4. Difference between encryption and decryption.

Encryption converts readable plaintext into ciphertext using an algorithm and key. Decryption converts ciphertext back into readable plaintext using the appropriate key.

### Q5. Explain symmetric and asymmetric cryptography and compare them.

Symmetric cryptography uses the same secret key for encryption and decryption. It is fast and suitable for large amounts of data, but the key must be shared securely. Examples: AES and DES. Asymmetric cryptography uses a public key and a private key. It is generally slower, but helps with secure key exchange and digital signatures. Examples: RSA and ECC. In practice, secure systems often combine both types.

### Q6. What is hashing, and how is it different from encryption?

Hashing converts data into a fixed-length digest and is designed to be one-way. Encryption protects data in a

reversible way, so it can be decrypted with the correct key. Hashing is commonly used to check integrity and store password verifiers.

### Q7. Why is achieving confidentiality, integrity, and non-repudiation together challenging?

These goals require different protections. Encryption supports confidentiality, hashes and message authentication codes help protect integrity, and digital signatures support non-repudiation. Managing keys correctly and avoiding implementation errors make all three difficult to guarantee together.

### Q8. Explain cryptography’s role in network security with examples.

Cryptography protects data sent across networks. Confidentiality: TLS encrypts web traffic so outsiders cannot easily read it. Integrity: a message authentication code can detect unauthorized changes. Authentication: certificates help a browser verify a website’s identity. Non-repudiation: a digital signature can provide evidence that a particular private key signed a document.

### Q9. Name two network security protocols that use cryptography.

TLS protects data exchanged between applications, such as a browser and website. IPsec protects IP network traffic and is often used in virtual private networks (VPNs).

### Q10. Discuss the evolution of cryptography and the impact of computing advances.

Early cryptography used simple substitution and transposition ciphers, which could often be broken by studying letter patterns. Computers enabled stronger, more complex algorithms, but also made large-scale codebreaking faster. Modern systems use algorithms such as AES and RSA and rely on sound key management. As computing power grows, old algorithms and short keys may become unsafe, so cryptographic standards must be updated.

### Q11. Trace cryptography from classical methods to modern systems.

Classical methods included the Caesar cipher and Vigenère cipher. During World War II, the Enigma machine was used to encrypt military communications, and codebreaking became strategically important. In the 1970s, DES became a widely used computer-era encryption standard. Public-key cryptography was publicly introduced in the 1970s, enabling key exchange and digital signatures. RSA followed in 1977, and AES was selected in 2001 to replace DES for many uses. Modern systems combine encryption, authentication, and digital signatures.

### Q12. Who introduced public-key cryptography, and in which year?

Whitfield Diffie and Martin Hellman publicly introduced the concept of public-key cryptography in 1976.